

Discovery 450, Optima 450w, Discovery 750, Discovery 750w, SIGNA Artist, and SIGNA Architect GOC Upgrade to GOC with Dell T5820 Host PC



5829488
Revision 3.0
US English
© 2020-2021 General Electric Company
All rights reserved.

Language Policy

Direction 2128126 - Language Policy For Service Documentation

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. - Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. - Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. - Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 - 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 - 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 - 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他形式的危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 - 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 - 請勿試圖維修本設備，除非您已查閱並瞭解本維修手冊。 - 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obsluženého personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAAR- SCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlke-teenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. - Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. - Εάν δεν λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ÆDVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. - Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. - Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. - Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.
ЗАСТЕРЕЖЕННЯ (UK)	Даний посібник з експлуатації доступний тільки англійською мовою. - Якщо постачальник послуг клієнта спілкується іноземною мовою, тоді клієнт зобов'язаний забезпечити переклад. - Заборонено проводити огляд обладнання без попереднього звертання до даного посібника з експлуатації і розуміння інформації, поданої у ньому. - Недотримання цього застереження може завдати шкоди здоров'ю постачальника послуг, оператора або пацієнта через ураження електричним струмом, механічну травму або інше ушкодження.

Revision History

Revision	Date	Description
----------	------	-------------

3.0	February 2021	Section 1: Reorganized Revision History and Getting Started to match current standards. Section 2.1: Removed 8400 from list of GOCs. Section 2.1: Updated Harvested Materials table. Section 3.2: Removed 8400 from list of GOCs. Section 3.2.1: Added upgrade cable maps. Section 3.3: Removed monitor information. Section 3.3: Corrected connections. Section 3.3: Moved iButton location.
2.0	December 19, 2019	Section 2.1.1: Updated the list of host PCs in the first paragraph. Section 2.1.3: Removed sentence about resolving surface coil compatibility issues. Section 2.1.5: Removed last paragraph about service and operator manuals being delivered with the Upgrade Kit. Sections 2.1.10.2: Removed all mentions of Harvest testing. Section 2.1.10.3: Removed all mentions of testing harvested components. Updated the list of host PCs in Table 3, Harvested Materials. Section 3.1: Updated the list of host PCs. Section 3.1.2: Updated Figure 2, GOC with Dell T5810. Section 3.1.3: Updated Figure 3, GOC with Dell T5820. Section 3.1.4: Updated Figure 4, Remove GOC covers and power cord, to not show an HDMI splitter. Added step 18 for disconnecting cable E3037 from the host computer. Section 3.1.5.1: Updated Figure 6, GOC cover removal and power cable installation, to not show an HDMI splitter. Updated step 4 to connect power cable E0003 to the GOC power cable adapter (5820948). Added step 5 to connect the power cable adapter (5820948) to PDU J1. Removed step 10 about Ethernet connections. Updated step 11 to connect Run E3503 to J7. Section 3.2: Removed the figure of the GOC with Dell T5820, since it already appears in section 3.1.3. Updated the note about GOC Ethernet ports to refer specifically to Site Network Ethernet port #2. Section 3.3: Updated step 2 to refer to part number 5818377. Section 4.1.2: Updated step 2 for clarification. Removed steps 5 and 6 about running a USB mouse fix. Routed in MyWorkshop as DOC2270584, Revision 6
1.0	June 30, 2019	Initial Release of part number 5829488. (Revision 1.0 is referenced in DOC2270584, rev 4, in MyWorkshop.)

Contents

Chapter 1 Getting Started.....	1
1.1 Introduction.....	1
1.2 General safety precautions.....	1
1.3 Material policy overview	2
1.3.1 Returned material.....	2
1.3.2 Disposal of material	2
1.3.3 Harvest test procedure.....	2
1.3.4 Harvest items.....	2
1.4 Product delivery instructions.....	3
1.5 Product locator.....	3
1.6 Damage in transportation.....	4
1.7 MR system shipping and storage requirements.....	5
1.8 Required tools and equipment.....	5
Chapter 2 GOC Upgrade Procedures.....	7
2.1 Preparing for the Global Operator Console (GOC) upgrade	7
2.2 Lock Out/Tag Out (LOTO) procedure for GOC upgrade.....	7
2.3 Global Operator Cabinet (GOC) upgrade to GOC with Dell T5820 host computer	7
2.3.1 Upgrade cable maps.....	7
2.3.2 Interconnect maps for GOC with Dell T5820.....	9
2.3.3 Removal procedure for the existing GOC	10
2.3.4 Install the new GOC with a Dell T5820 host PC.....	12
2.3.4.1 Connect the GOC module cables	13
2.3.4.2 Connect system cables to the GOC	16
2.3.4.3 Connect GOC PC Cables for Dell T5820	16
2.3.4.4 Restoring the Global Operator Cabinet (GOC) covers.....	18
Chapter 3 Power On Procedures	21
3.1 Power on procedures.....	21

Page intentionally left blank

Chapter 1 Getting Started

1.1 Introduction

This publication provides instructions to upgrade an existing GOC with an HP Z400, HP Z420, or a Dell T5810 Host PC to a GOC with a Dell T5820.

To upgrade to a GOC with the Dell T5820 host PC, the entire GOC and, if required, the LCD monitor must be replaced.

Follow all other instructions as provided in the procedures in this manual to ensure satisfactory completion of Harvest and Product locator activities.

WARNING



FERROUS MATERIAL HAZARD

Delivery equipment and other tools and parts required for this installation contain ferrous material. Ferrous materials can become dangerous projectiles within the magnetic field.

Keep all ferrous tools at least 10 feet away from the magnet.

Site Ready Check

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Pre-Delivery FDO Checks

Before the shipment is made, be sure that FDO review has been completed by consultation with the Upgrade Specialist.

Upgrade Schedule Considerations

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer.

Upgrade Components

GOC with Dell T5820 Host PC

1.2 General safety precautions

Safety

Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must:

- Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735).
- Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses).
- Comply with all site-specific training and workplace safety requirements.

If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

1.3 Material policy overview

1.3.1 Returned material

Based on GE Healthcare Policy 3.9, all requested material is to be promptly returned as directed, intact to either GoldSeal or Approved GE Recycling Center. Do not remove or swap any parts from the material to be returned.

1.3.2 Disposal of material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. To make sure you are compliant with all local, state, or national environment, health, and safety regulations, dispose of all items based on the local GEMS Policy and Procedures, and other applicable published guidance. Contact the GE Recycling Center for assistance.

1.3.3 Harvest test procedure

NOTICE

RISK OF DAMAGE

Equipment may be damaged prior to removal.

The equipment must be tested prior to the system shutdown/upgrade to allow appropriate harvest capabilities.

Before going on-site:

- Print the proper regional address labeling from *Harvest Return Addresses* (DOC2100599).
- Make sure you can download the appropriate *System Harvest Test Procedure* from the online documentation library using:
 - DOC1899393 for MR450, MR450W, MR750, MR750W, Artist, and Architect
- If you have not been contacted to test the system or cannot download the document, contact the local harvest team. For all other requests, contact the @HEALTH Global Harvest Team via email: GlobalHarvestTeam@ge.com.

After testing is complete, submit the test record for review. After these test records are reviewed, they will be processed for electronic signatures.

Package the equipment and follow your region's method of returning harvest equipment.

NOTE

Where possible, use packaging from the new upgraded component for the equivalent harvested parts.

1.3.4 Harvest items

Return the following items for harvest after deinstallation.

Table 1-1 Harvest Items

Description	Qty
MR Host PC (HP Z400, HP Z420, or Dell T5810)	1
GOC PDU assembly	1
GOC audio box	1
8 port switch kit with 4 cushions	1

1.4 Product delivery instructions

- The *Shipping Document* lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if catalogs and/or packing boxes are missing and/or noted as shipped short.
- NOTE**
For full systems and large components, two or more people are required to accept delivery of equipment due to weight and size.
- Labels that are attached to the outside of packing boxes summarize box contents.

Product Delivery Instructions (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

1.5 Product locator

During installation, the Install FE shall confirm that the model and serial numbers are part of the asset. Use either the gib web interface (http://gib.gehealthcare.com/gib/gib_entry.jsp) or the MyPLCReader app. This information will then be loaded directly to Siebel by the IB verification team. If components are missing, submit a Global Installation Quality (GIQ) request.

To install the MyPLCReader app, use the following procedure.

1. From the GE app icon, search for and Install the MyPLCReader app on your iPhone or iPad.

Figure 1-1 MyPLCReader app

2. Update the Configuration page.
 - a. The first time the app opens, the Configuration page will open.
 - b. Enter your SSO, Country and **if in US select the edit button to change the default email address** (by default, this app will send the email to the Europe IB Team Resource.EuropeanCentralAdminGIB@ge.com). The email to be used in US is usa.ib.verification@ge.com.
3. Start your first scan.
 - a. Go to the Menu, make the appropriate selection for the job you are performing: **New Install**, **Upgrade System**, or **Update System**.
 - b. Enter the applicable data, install end date, system ID, and RFS number.
 - c. Click **Scan Bar Codes** and start scanning each bar code on the PL Cards.
 - d. Click the **Email** button.
 - e. The email is sent to the default email address, which was set up earlier.

1.6 Damage in transportation

Examine all packages closely at delivery. If damage is apparent, do the following:

1. Make sure the notation *damage in shipment* is written on all copies of the freight or express bill before delivery is accepted or signed for by a General Electric representative or a hospital receiving agent.
2. Whether noted or concealed, you must report damage to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and hold the contents and containers for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

1.7 MR system shipping and storage requirements

Environmental requirements for shipping and storage.

NOTE

Some content may not apply to your components.

1. The customer must provide a shipping dock or other staging area for unloading system components for full systems and large components (magnet, cabinets, patient table, and so on).
2. The customer is responsible for unloading and moving all system components from the delivery truck or van to final locations in the MR suite. This includes:
 - Door and other openings are sufficiently wide to allow passage
 - Sufficient space is provided for any required dollies or rigging
3. MR systems and components must be shipped and stored within the environmental requirements as listed in [Table 1-2 Transportation and storage environmental conditions on page 5](#).

Table 1-2 Transportation and storage environmental conditions

System equipment	Temperature range	Temperature change	Relative humidity	Humidity change
Electronic Cabinets and Equipment	-22 to 140°F (-30 to 60°C)	68°F/Hr (20°C/Hr)	10-80% non-condensing	30%/Hr
Surface Coil	-22 to 122°F (-30 to 50°C)	68°F/Hr (20°C/Hr)	10-95% non-condensing	30%/Hr

1.8 Required tools and equipment

DANGER



FERROUS MATERIAL HAZARD

If the magnet is energized, some tools and parts required for this upgrade may contain ferrous material that will be strongly attracted to the magnet and may become dangerous projectiles.

Keep all ferrous tools as far from the magnet as possible.

NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

Table 1-3 General PPE required gear

Item	Description
1	Composite safety shoes for foot protection.
2	Gloves for hand protection.
3	Safety glasses for eye protection.

Tools to install/upgrade the various components in the system are listed below:

Table 1-4 Required tools and equipment

GE Part Number	Description
5112581	Nonmagnetic Titanium Service Tool Kit, Large Set
46-271138G1	Restricted Access Control Kit (contains two plastic warning signs for posting at site during installation and service activity)
46-198094P1	Wrist Grounding Strap
-	Flashlight
-	Assorted Cable Ties
-	Service Methods

Chapter 2 GOC Upgrade Procedures

2.1 Preparing for the Global Operator Console (GOC) upgrade

About this task

The following steps must be performed prior to removing the GOC.

Procedure

1. Ask the customer to archive all images to an external storage device such as a PACS or DVDs.
2. Make sure local network information is noted for reconnection to the local network.
3. Perform a Save Info.
4. Perform a disk wipe on all applicable hard drives in the old PC. See latest revision of the *Disk Management Tool Service Manual* (5500610-1EN).

2.2 Lock Out/Tag Out (LOTO) procedure for GOC upgrade

Procedure

1. Confirm that all customer and patient data has been archived or stored to a separate network server. All existing system information will be lost as a result of this upgrade.
2. Perform a Hello scan on the system.
3. Shut down the GOC.
4. Perform the LOTO procedure for the GOC at the PGR cabinet. See the section on LOTO Global Operator Console / Operator Workspace in the *MR Service Safety Manual* (5452735).

2.3 Global Operator Cabinet (GOC) upgrade to GOC with Dell T5820 host computer

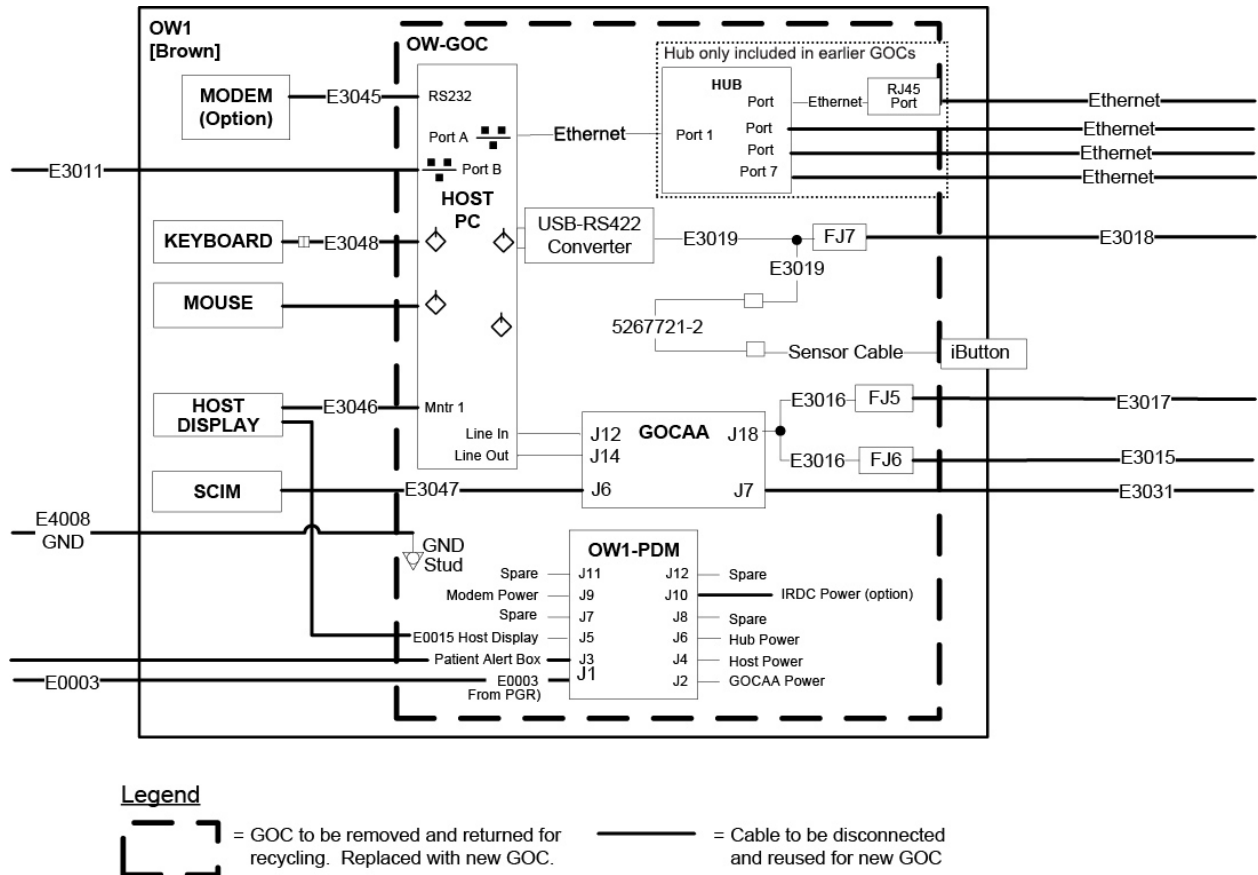
The following upgrade procedures provide instructions for upgrading an existing GOC with an HP Z400, HP Z420, or a Dell T5810 PC to a GOC with a Dell T5820 PC.

2.3.1 Upgrade cable maps

Provides information on cable reuse and component transfer from the old GOC to the new GOC.

Cable and component upgrade from a GOC with an HP z400, HP z420, or a Dell T5810

Figure 2-1 Systems without IRD

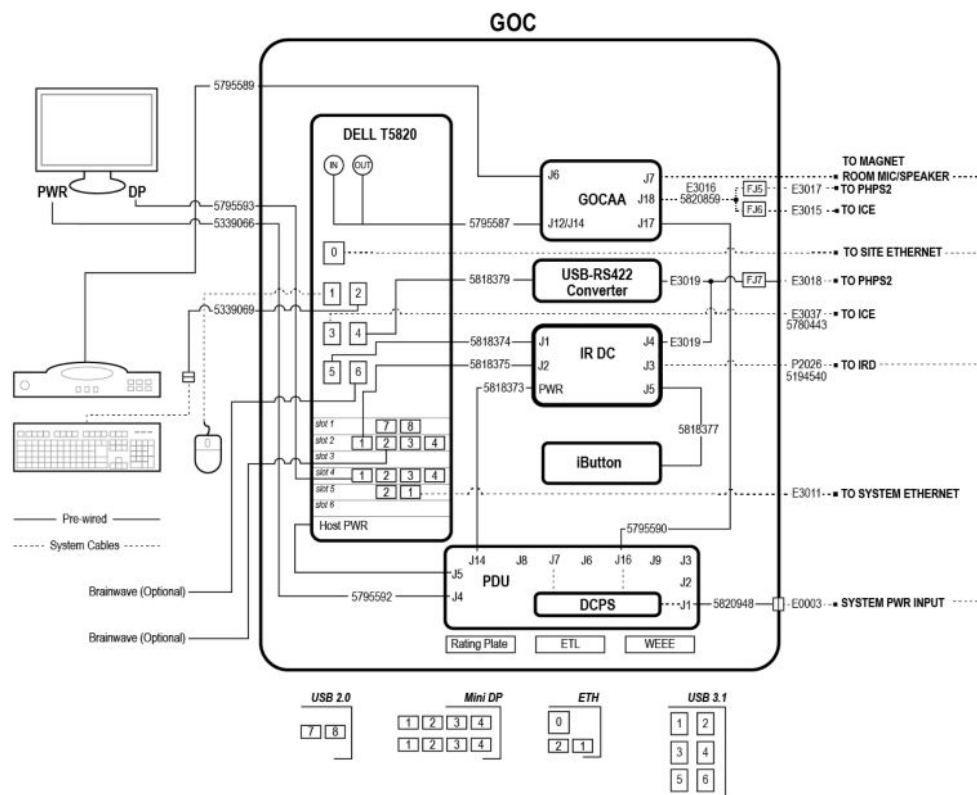


2.3.2 Interconnect maps for GOC with Dell T5820



Cable and component upgrade for a GOC with a Dell T5820

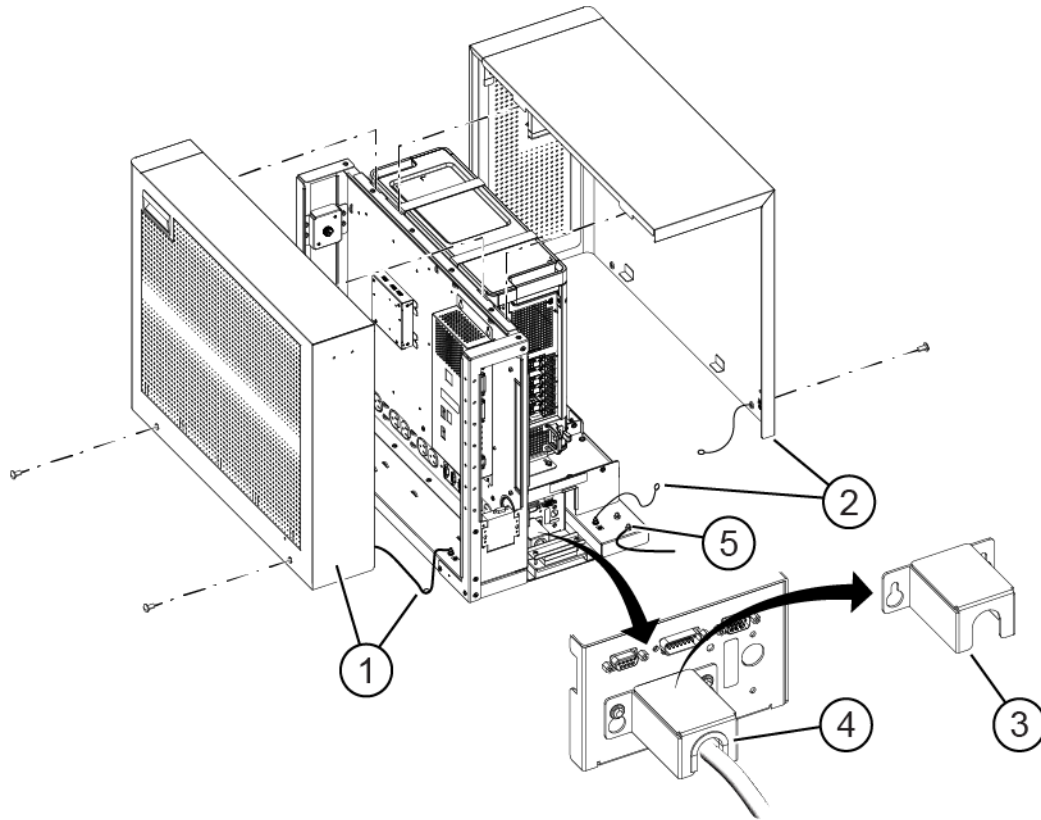
Figure 2-3 GOC with Dell T5820



2.3.3 Removal procedure for the existing GOC

Cables routed to GOC will be reused for connecting to new GOC. Remove cable ties as necessary to make it easier to remove the cables that are routed into the GOC.

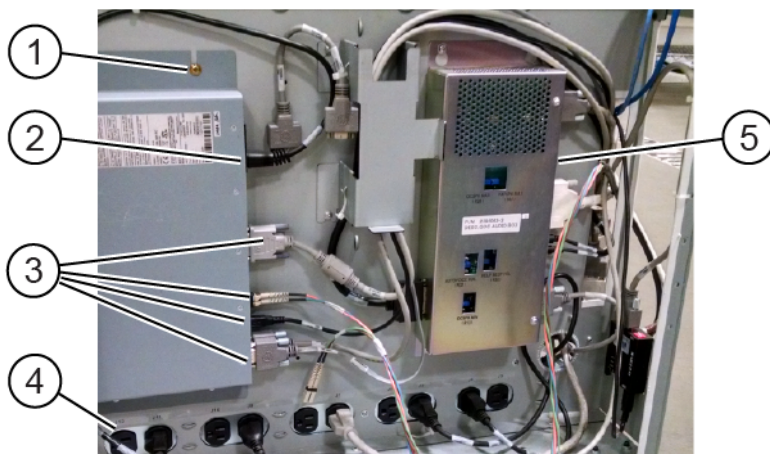
Figure 2-4 Remove GOC covers and power cord



Item	Description
1	Right side cover and ground cable
2	Left side cover and ground cable
3	Power cable cover removed
4	Power cable installed
5	GOC ground cable

1. For right side cover removal:
 - a. Remove two screws.
 - b. To access cover ground cable, carefully move cover forward and then lift up and to the right.
 - c. Disconnect ground cable and remove the cover.
2. For left side cover: Remove the two screws and lift up. Disconnect the ground cable and remove cover.
3. Loosen screws and remove power cable cover.
4. Loosen power cable clamp. Disconnect power cable from J1.
5. Disconnect ground cable.

Figure 2-5 GOC module cable removal



Item	Description
1	M5 (4) screws attaching the IRD module
2	IRD power cable
3	IRD cables
4	Power cables
5	Runs E3047 and E3031

6. Disconnect power cables from GOC PDU.

NOTE

If IRD Fiber Optic Converter Box option is installed, complete the next three steps.

7. Disconnect all data cables from IRD module.
8. Disconnect the power cable from IRD module.
9. Remove the four M5 screws attaching the IRD module to GOC and remove module. The module will be transferred to the new GOC.
10. Label and disconnect all Ethernet cables routed to the LAN Hub.
11. Disconnect Runs E3047 and E3031 from J6 and J7 from the GOC AA module.
12. Disconnect all cables from the USB ports.
13. Disconnect the mouse, SCIM, and keyboard to be reused.
14. Disconnect the monitor.
15. Disconnect cable run E3017 (from the GOC) from E3016/FJ5 (from the PGR cabinet).
16. Disconnect cable run E3015 (from the GOC) from E3016/FJ6 (from the PGR cabinet).
17. Disconnect cable from GOC to E3018 (from the PGR cabinet).

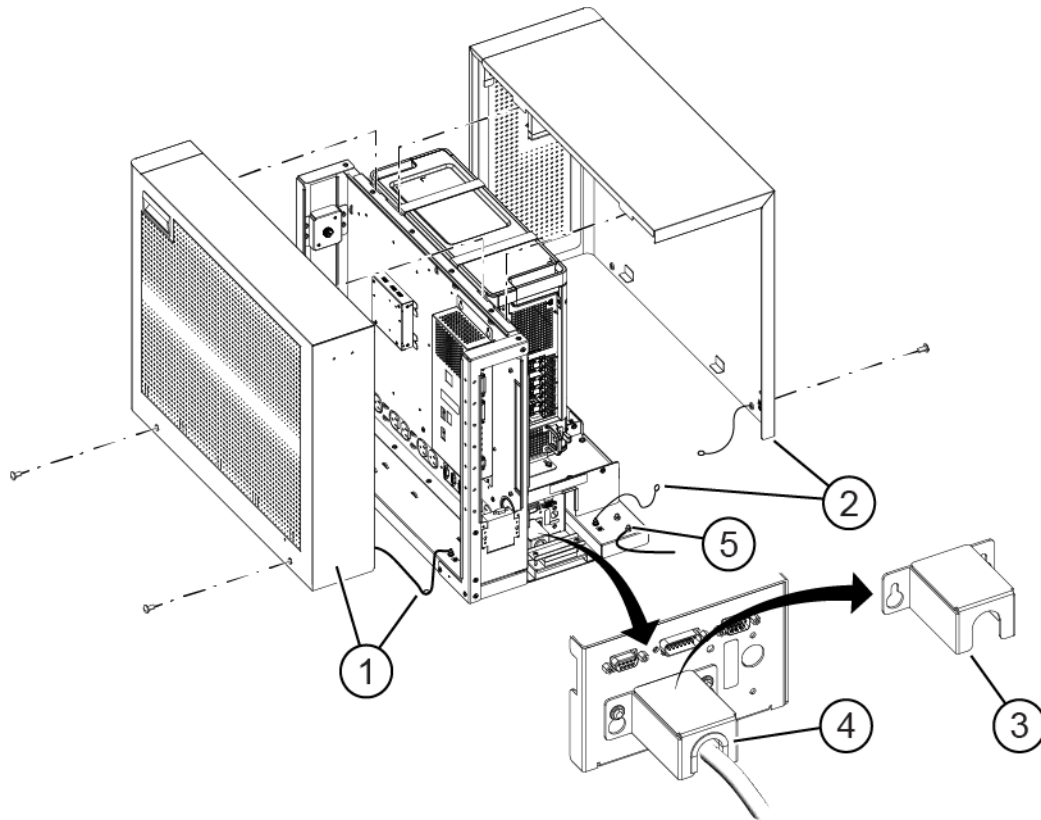
Attach covers to the GOC and prepare to return the GOC for harvest using the packaging material from the new GOC. The GOC must be returned for harvest in its entirety with all subcomponents. The only exception is the hard drive, which may be removed from the Host PC and given to the customer.

2.3.4 Install the new GOC with a Dell T5820 host PC

Carefully unpackage the new GOC and move into place. The packaging materials will be used for return of the old GOC.

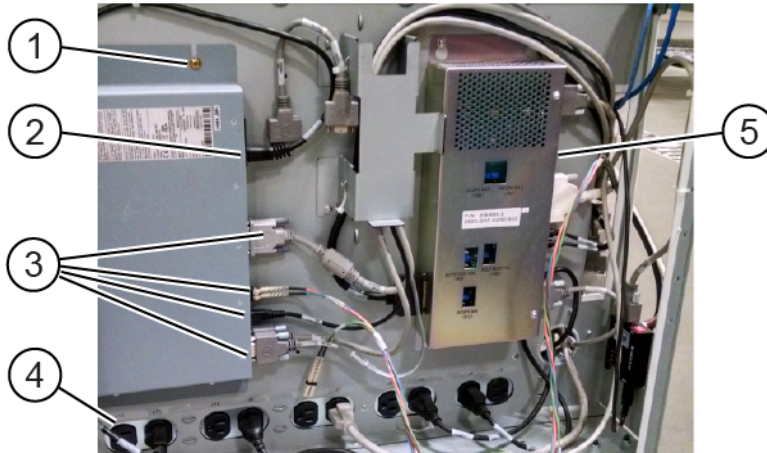
2.3.4.1 Connect the GOC module cables

Figure 2-6 GOC cover removal and power cable installation



1. For right side cover removal:
 - a. Remove two screws.
 - b. To access cover ground cable, carefully move cover forward and then lift up and to the right.
 - c. Disconnect ground cable and remove the cover.
2. For left side cover: Remove the two screws and lift up. Disconnect the ground cable and remove cover.
3. Loosen screws and remove power cable cover.
4. Loosen power cable clamp. Connect 5820948 to PDU-J1.
5. Route and connect existing power cable E0003 to E0500 (5820948).
6. Connect Run 4008 to GND. Make sure the nut securing ground cable is firmly tightened to GND stud. Tighten cable clamp to secure cables in place.

Figure 2-7 Cable connections to the GOC modules



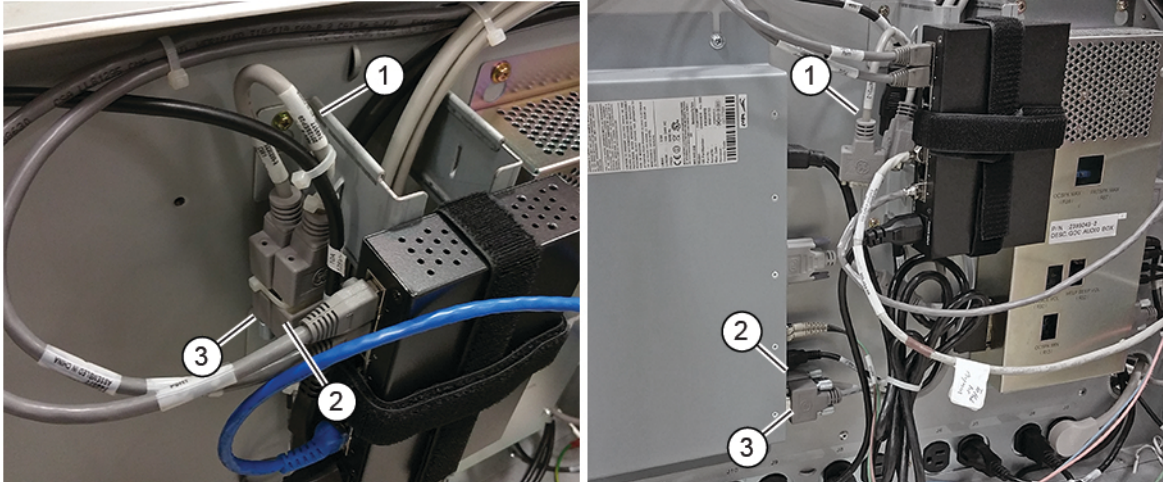
Item	Description
1	M5 screws attaching the IRD module
2	IRD power cable
3	IRD cables
4	Power cables
5	Runs E3047 and E3031

NOTE

Complete the next four steps if:

- IRD Fiber Optic Converter Box option is being transferred from removed GOC
- OR
- Site is being upgraded to IRD from an LED Display.

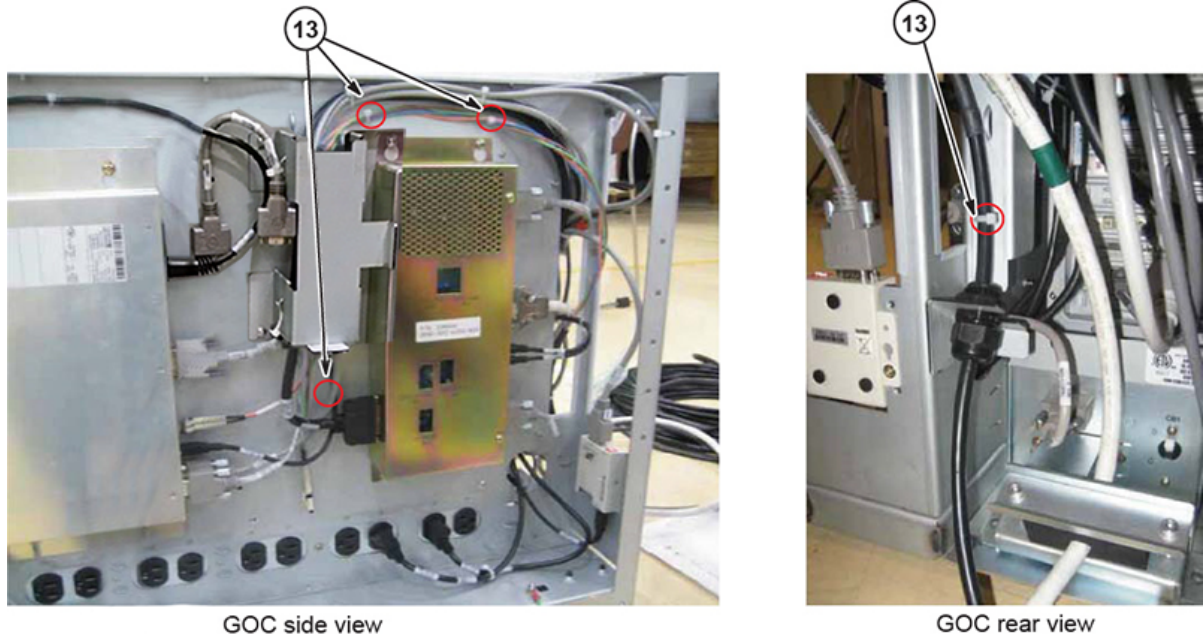
7. Disconnect both ends of jumper cable 5267721-2. This jumper cable will no longer be used. Connect the free end of cable run E3019 to IRDC J4 and the free end of the sensor cable to IRDC J5.

Figure 2-8 Jumper cable before (left) and after (right)

1	Jumper cable
2	Cable Run E3019 (to IRD J4)
3	Sensor cable (to IRD J5)

8. Install the IRD Fiber Optic Converter Box using the four M5 screws.
9. Connect the power cable to the IRD module.
10. Connect the following cables to the IRD Fiber Optic Converter Box.
 - a. Connect the DVI-D cable from the Host PC to J2.
 - b. Route fiber optic cable Run P2026 to GOC and connect to J3.
 - c. Connect the USB cable from the Host PC to J1.
 - d. Connect Run E3019 to J4 and the Sensor cable to J5.
11. Connect Ethernet cables directly to the Host PC. If multiple Ethernet connections are required, use a customer-supplied Ethernet switch.
12. Reconnect Run E3047 to J6 and Run E3031 to J7 on the GOC AA.
13. Reconnect E3047, the mouse, and the keyboard to the SCIM if previously disconnected.
14. Reconnect the monitor unless the monitor will be replaced.
15. Reconnect power cables to PDU Module.

Figure 2-9 Securing the fiber optic cable



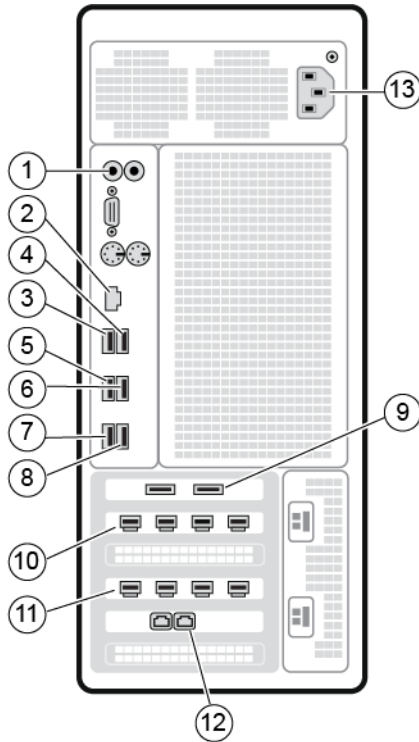
16. Secure the fiber optic cable with tie wraps at points indicated and make sure the tie wraps are loose as to not damage the optical cable.

2.3.4.2 Connect system cables to the GOC

1. Connect cable run E3017 (from the GOC) to E3016/FJ5 (from the PGR cabinet).
2. Connect cable run E3015 (from the GOC) to E3016/FJ6 (from the PGR cabinet).
3. Reconnect all Ethernet cables.

2.3.4.3 Connect GOC PC Cables for Dell T5820

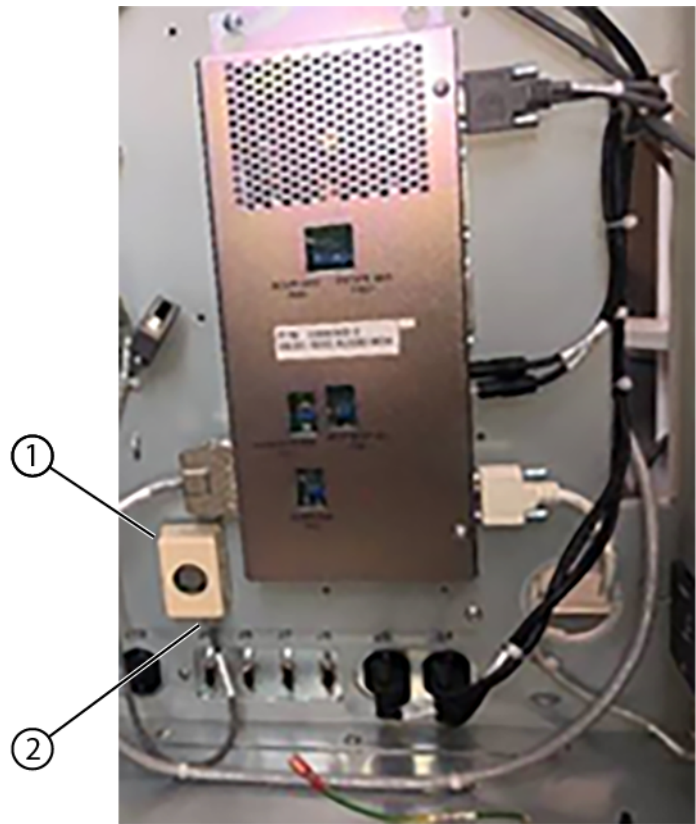
The connection points of cables routed to the PC are described as follows. Several of the cables shown are delivered with the GOC and are preconnected.

Figure 2-10 PC area connections (Dell T5820)**Table 2-1 Dell T5820 cable configuration**

Item	Run #	Description
1	NA	Line In/Out cable to GOCAA J12/14
2	NA	Site network Ethernet cable
3	NA	Mouse USB cable
4	E3048	Keyboard USB extender cable to keyboard USB cable
5	E3037	ICE USB extender cable
6	NA	USB-RS422 Converter USB cable to USB-RS422 Converter
7	NA	IRDC USB cable to IRDC J1
8	NA	NA
9	NA	NA
10	NA	IRDC miniDP cable to IRDC J2
11	E3046	Host LCD Monitor miniDP cable to Host LCD Monitor
12	E3011	ETH1 (right port) cable to PGR J2
13	NA	Power cable

1. Attach the iButton sensor assembly to the GOC using Dual Lock Tape, 400 Stems Per Square Inch (2341779) and Dual Lock Tape, 170 Stems Per Square Inch (2341779-2).

Figure 2-11 GOC iButton



1	iButton
2	iButton connection

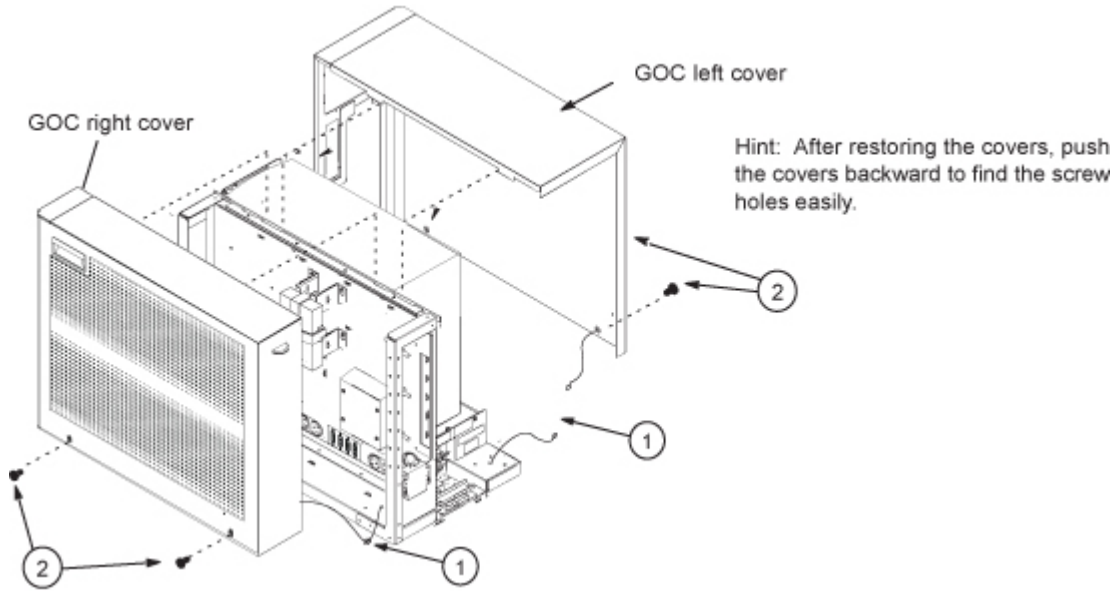
2. Make sure the IRDC J5 to iButton cable (5818377) is connected to the iButton sensor.

2.3.4.4 Restoring the Global Operator Cabinet (GOC) covers

Procedure

1. Connect ground cables and install both side covers by inserting tabs at top of covers into slots.
2. Install screws that were removed in earlier procedure.

Figure 2-12 Restore GOC covers



Page intentionally left blank

Chapter 3 Power On Procedures

3.1 Power on procedures

Remove LOTO

To remove LOTO, see the section on LOTO Global Operator Console / Operator Workspace in the *MR Service Safety Manual* (5452735).

Finalization procedures

1. Power on Host PC.
2. The host PC comes with preloaded software. Doing the Software Load procedure is recommended; refer to the appropriate Service Methods Documentation to do so.
3. Update eLicense site with new Host ID and Secure Service Access (SSA) site with new Service License ID.

Page intentionally left blank

